Laplace Domain

Control Engineering

Course Schedule

1. Systems
2. Continuous-time Controllers
3. Laplace Domain
4. Poles/Zeros & Stability
5. Controller Design
6. Discrete-time Controllers
7. State Space Representation
8. Advanced Control Methods

Laplace Domain

- Laplace transform
- Transfer function
- Frequency response
- Bode plots
- Bandwidth

Why the Laplace Transform?

- Converts differential equations (dynamic systems) into algebraic equations → easier to solve

- Simplifies analysis and design

- Natural language of control theory

Reminder: Fourier Transform

Transformation from time to frequency domain:

$$\mathcal{F}\{f(t)\} = F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}dt$$

Tells which frequencies ω are present in a signal or system

To be exact, tabulation of both magnitude and phase response at each frequency → complex numbers:

$$F(\omega) = |F(\omega)|e^{i\phi}$$

- $|F(\omega)|$: how much of that frequency in the signal (how much the system scales each frequency)
- ϕ: time shift of that frequency (how much the system shifts each frequency)

Idea of Laplace Transform

Extend Fourier transform to include information about growth/decay behavior

→ Add exponential weighting:

$$e^{-i\omega t} \rightarrow e^{-i\omega t} \cdot e^{-\sigma t}$$

oscillation growth/decay

Combine frequency ω and growth/decay rate σ:

$$\omega \rightarrow s = \sigma + j\omega$$

Definition of Laplace Transform

Integral transform from time to complex frequency s:

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty f(t)e^{-st}\,dt$$

$$s = \sigma + j\omega$$

Back transformation:

$$\mathcal{L}^{-1}\{F(s)\} = f(t) = \frac{1}{2\pi j}\int_{\sigma-\infty}^{\sigma+\infty} F(s)e^{st}\,ds$$

Example

Laplace transform of t:

$$\mathcal{L}\{t\} = \int_0^\infty t \cdot e^{-st}\, dt$$

Using partial integration: $\int u\, dv = uv - \int v\ du$

$$u = t \quad \Rightarrow \quad du = dt$$

$$dv = e^{-st}\, dt \quad \Rightarrow \quad v = -\frac{1}{s} e^{-st}$$

$$\Rightarrow \int_0^\infty t \cdot e^{-st}\, dt = \underbrace{\left[-\frac{t}{s} e^{-st}\right]_0^\infty}_{=0} + \int_0^\infty \frac{1}{s} e^{-st} dt = -\frac{1}{s^2}\left[e^{-st}\right]_0^\infty = \frac{1}{s^2}$$

Laplace of Common Signals

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

Function	Time domain	Laplace domain
Unit Step	$u(t)$	$\frac{1}{s}$
Delayed unit step	$u(t-\tau)$	$\frac{1}{s}e^{-\tau s}$
Unit impulse	$\delta(t)$	1
Delayed unit impulse	$\delta(t-\tau)$	$e^{-\tau s}$
Ramp	$t \cdot u(t)$	$\frac{1}{s^2}$
Exponential decay	$e^{-\alpha t} \cdot u(t)$	$\frac{1}{s+\alpha}$
Sine	$\sin \omega t \cdot u(t)$	$\frac{\omega}{s^2+\omega^2}$
Cosine	$\cos \omega t \cdot u(t)$	$\frac{s}{s^2+\omega^2}$

Properties of Laplace Transform

Linearity:	$\mathcal{L}\{a \cdot f_1(t) + b \cdot f_2(t)\} = a \cdot F_1(s) + b \cdot F_2(s)$
Convolution:	$\mathcal{L}\{f_1 * f_2\} = \mathcal{L}\left\{\int_0^t f_1(\tau) \cdot f_2(t-\tau)d\tau\right\} = F_1(s) \cdot F_2(s)$
Derivative:	$\mathcal{L}\left\{\frac{df(t)}{dx}\right\} = sF(s)$
Integration:	$\mathcal{L}\left\{\int_0^t f(\tau)d\tau\right\} = \mathcal{L}\{\,(u * f)(t)\} = \frac{1}{s}F(s)$
Frequency Shifting:	$\mathcal{L}\{e^{at} \cdot f(t)\} = F(s-a)$
Time Shifting:	$\mathcal{L}\{f(t-a) \cdot u(t-a)\} = e^{-as} \cdot F(s)$
Time Scaling:	$\mathcal{L}\{f(at)\} = \frac{1}{a}F\left(\frac{s}{a}\right)$

From Differential Equation to Transfer Function

Example: PT2 system

Time domain:

$$u(t) \longrightarrow \boxed{\ddot{y}(t) + 2\zeta\omega_n\dot{y}(t) + \omega_n^2 y(t) = \omega_n^2 K u(t)} \longrightarrow y(t)$$

Frequency domain:

$$U(s) \longrightarrow \boxed{s^2 Y(s) + 2\zeta\omega_n s Y(s) + \omega_n^2 Y(s) = \omega_n^2 K U(s)} \longrightarrow Y(s)$$

Transfer function: $G(s) = \frac{Y(s)}{U(s)} = \frac{\omega_n^2 K}{s^2 + 2\zeta\omega_n s + \omega_n^2}$ (assuming zero initial conditions)

Transfer Function

Ratio of output $Y(s)$ to input $U(s)$ in Laplace domain

Fully describes input–output dynamics of system, independent of input

$$Y(s) = G(s) \cdot U(s)$$

In time domain, this corresponds to the convolution of the input with the impulse response of the system, **assuming zero initial conditions**:

$$y(t) = \int_0^t g(t-\tau) \cdot u(\tau) d\tau$$

Transfer Function for PT1 System

Differential equation:

$$\tau \cdot \dot{y}(t) + y(t) = K \cdot u(t)$$

Laplace transform:

$$\tau \cdot s \cdot Y(s) + Y(s) = K \cdot U(s)$$

Transfer Function:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{K}{1+\tau s}$$

independent of input

Example: Free Fall with Viscous Damping

$$m \cdot \dot{v}(t) + c \cdot v(t) = m \cdot g$$

where $u(t) = \begin{cases} 0, & t < 0 \\ m \cdot g, & t \geq 0 \end{cases}$ → $U(s) = m \cdot g \cdot \frac{1}{s}$

$$m \cdot s \cdot V(s) + c \cdot V(s) = m \cdot g \cdot \frac{1}{s}$$

$$V(s) = \frac{m \cdot g}{s(m \cdot s + c)} = \frac{g}{s\left(s + \frac{c}{m}\right)}$$

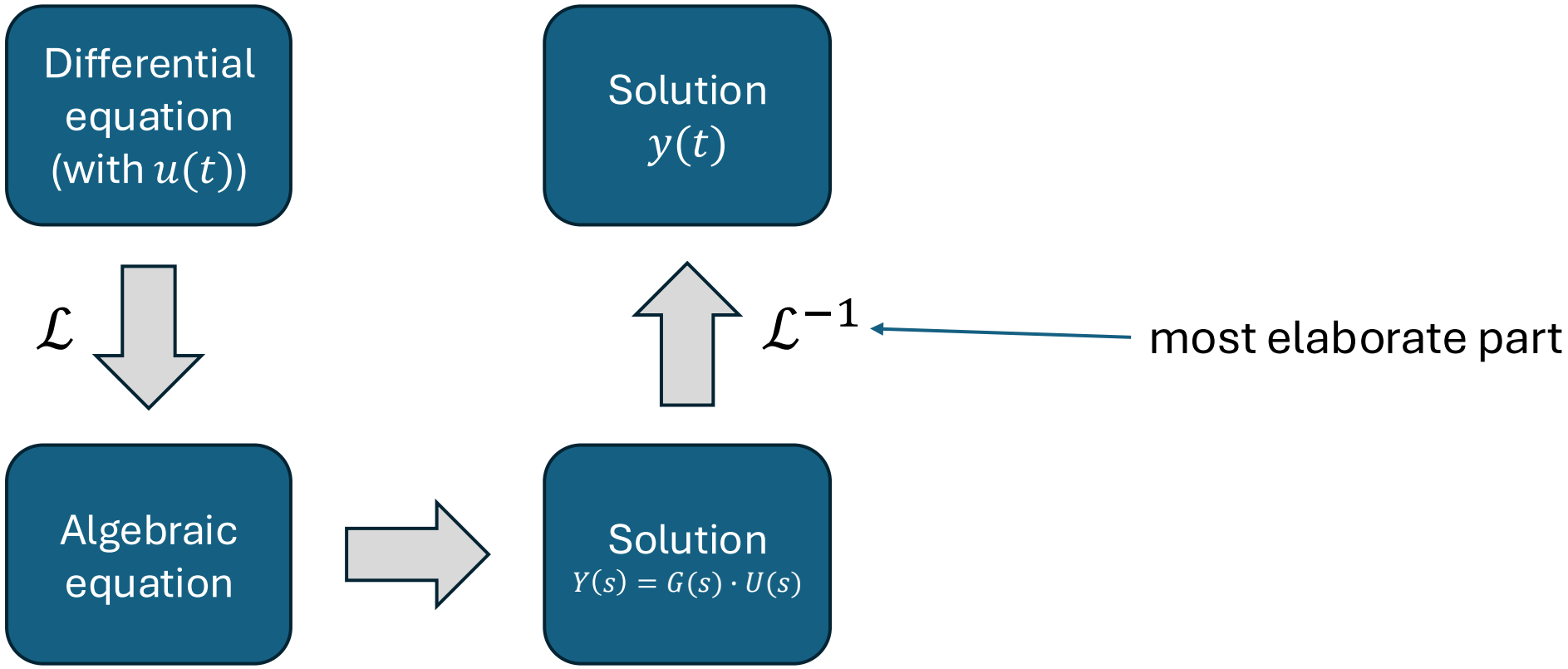
Transformation table:

$$\frac{\alpha}{s(s+\alpha)} \mapsto 1 - e^{-\alpha t}$$

$$\rightarrow v(t) = g \cdot \frac{m}{c} \cdot \left(1 - e^{-\frac{c}{m}t}\right)$$

$$\tau = K = \frac{m}{c}$$

System Response with Transfer Function



Transfer Function in Polynomial Form

$$G(s) = \frac{K}{1+\tau s}, \qquad G(s) = \frac{K}{\frac{1}{\omega_n^2}s^2 + \frac{2\zeta}{\omega_n}s + 1}, \qquad \ldots$$

Input coupling:
describes how the input enters the system (can contain time derivatives)

$$\rightarrow G(s) = \frac{b_m s^m + \cdots + b_1 s + b_0}{a_n s^n + \cdots + a_1 s + a_0} = \frac{b_0}{a_0}\frac{\frac{b_m}{b_0}s^m + \cdots + \frac{b_1}{b_0}s + 1}{\frac{a_n}{a_0}s^n + \cdots + \frac{a}{a_0}s + 1}$$

$$K = \frac{b_0}{a_0}$$

Intrinsic system dynamics:
characteristic polynomial of the system

fundamental theorem of algebra

$$\rightarrow G(s) = \frac{b_m}{a_n}\frac{\prod_{i=1}^{m}(s - s_{0i})}{\prod_{i=1}^{n}(s - s_{pi})}$$

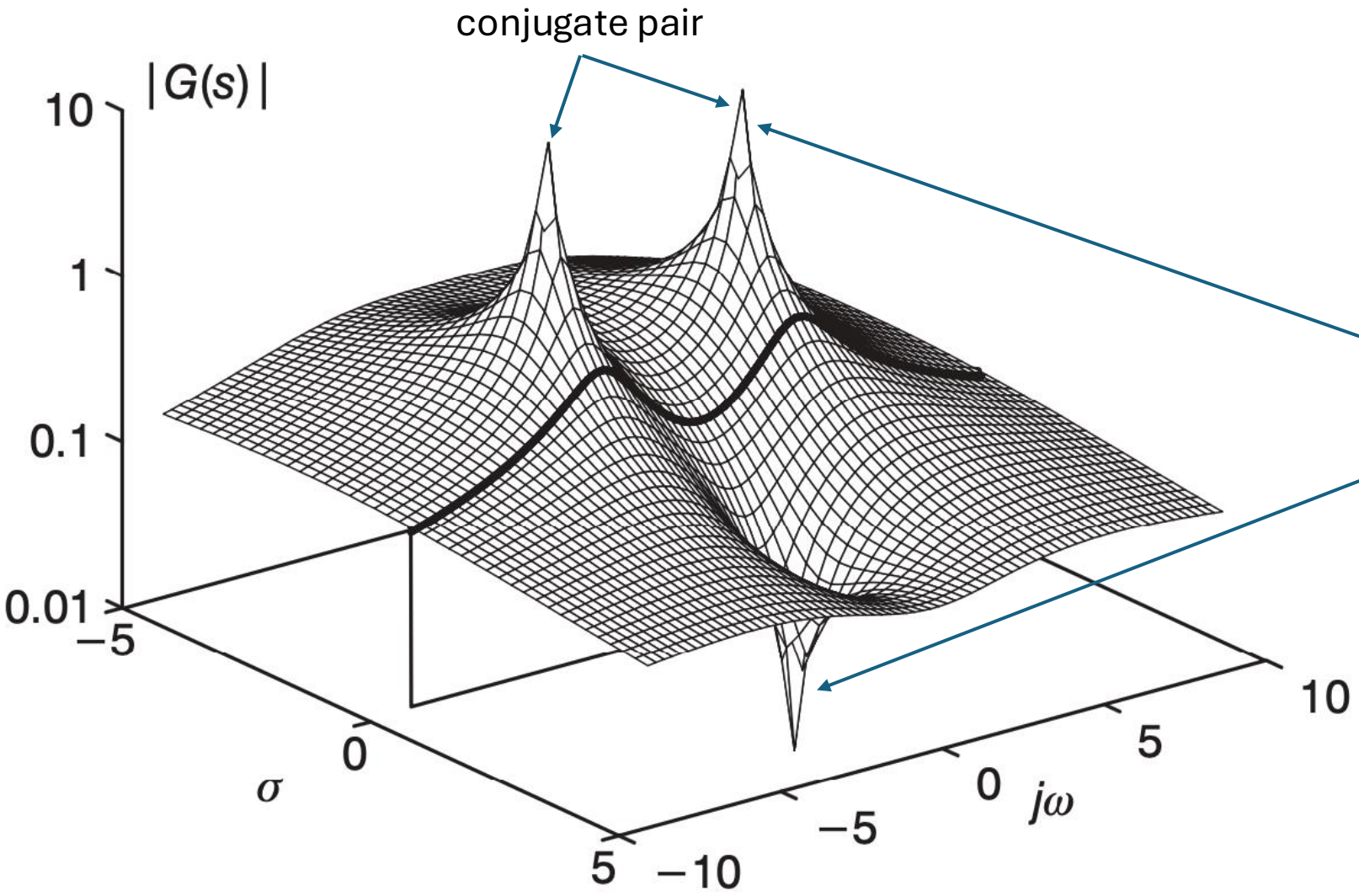
$$k = \frac{b_m}{a_n}$$

roots of numerator: m zeros s_0

roots of denominator: n poles s_p

Causality requires $n \geq m$ (output cannot depend on derivative of input)

Graphical Interpretation of Transfer Function



Contains (complex) poles and zeros

Poles and Zeros

PT1: one pole
PT2: two poles

Poles and zeros of a transfer function can be real or complex

Complex poles/zeros always appear in conjugate pairs for real physical systems (need for real coefficients in differential equations)

$$\boxed{\big(s-(\sigma+j\omega)\big)\big(s-(\sigma-j\omega)\big)=(s-\sigma)^2+\omega^2}$$

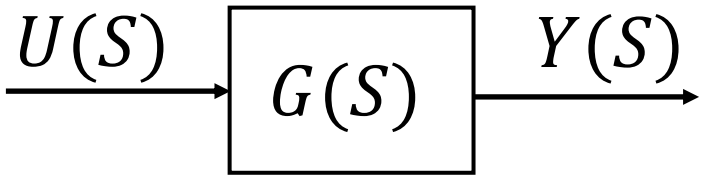
Poles determine stability and dynamics (system's natural response)

$y_i(t) = C_i e^{s_{pi} t}, \qquad y_{\text{hom}}(t) = \sum_i y_i(t)$

More about this later …

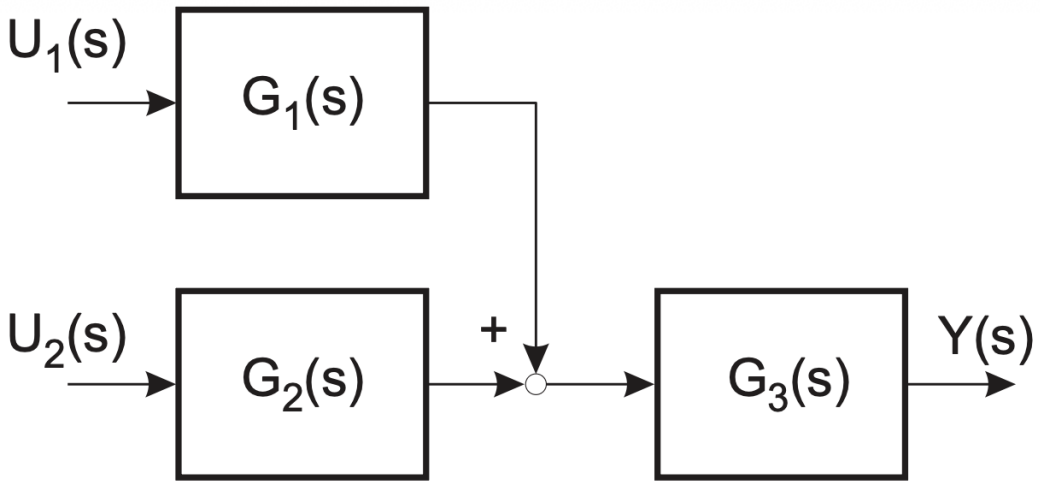
Zeros shape the forced (input-driven) response

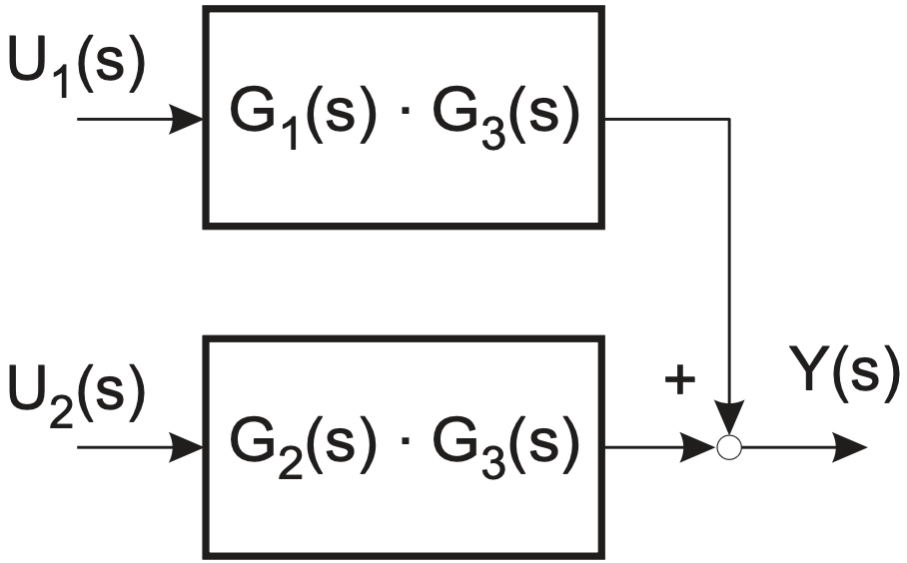
Block Diagrams

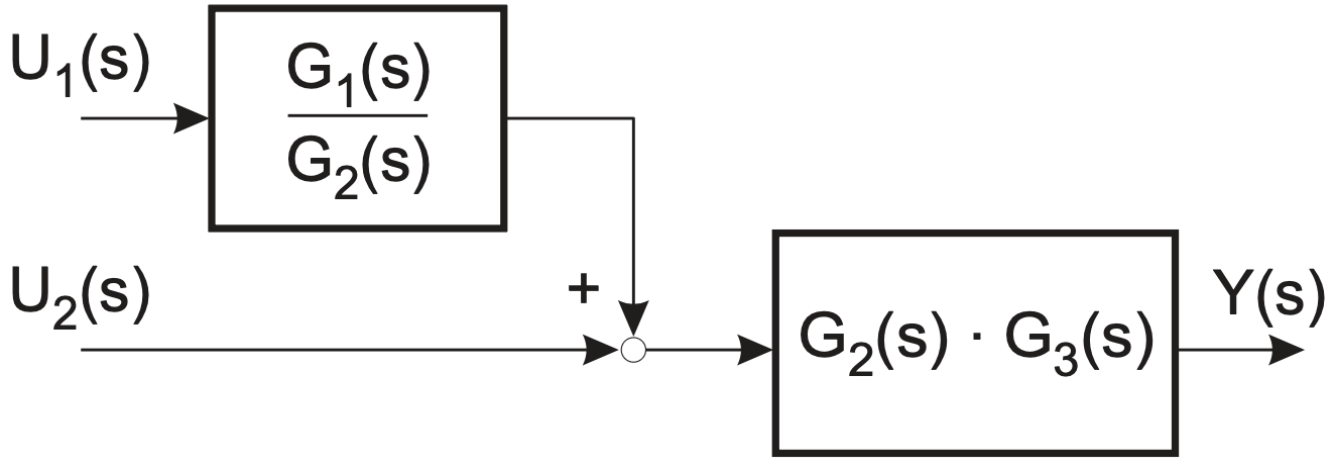


Modular representation

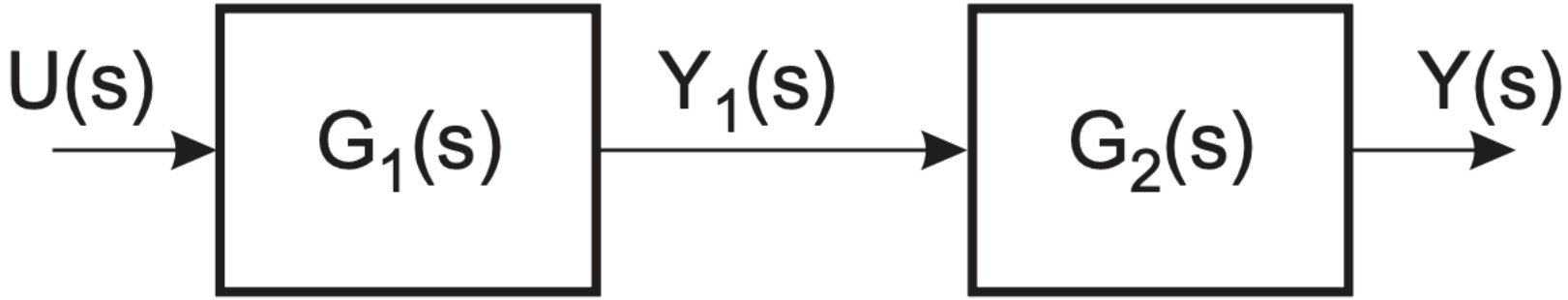
Block Diagram Algebra

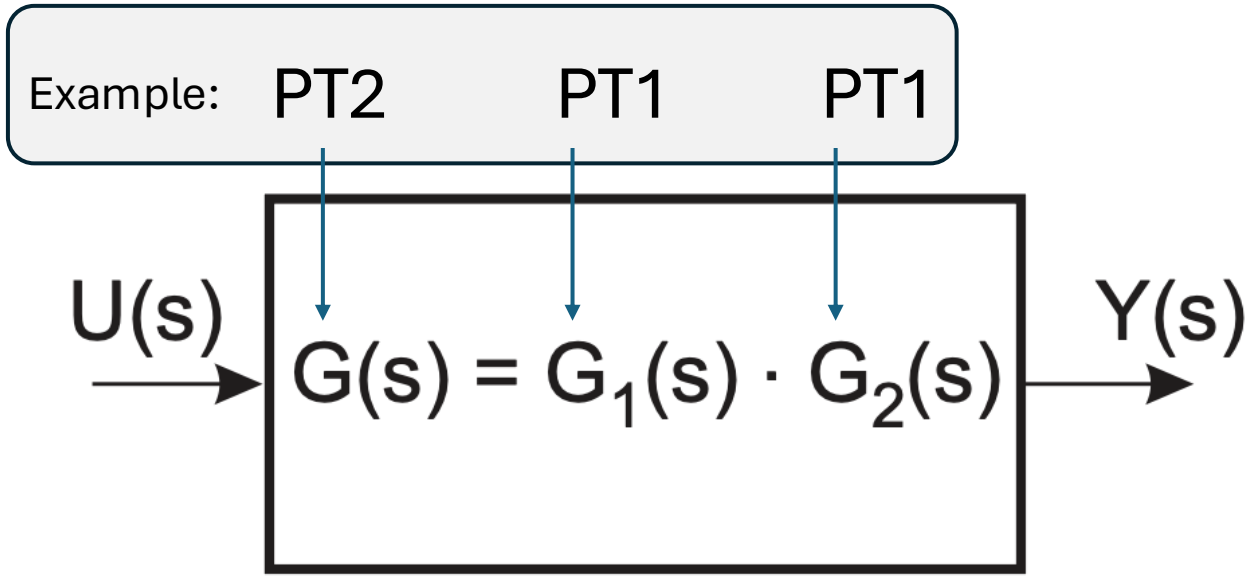




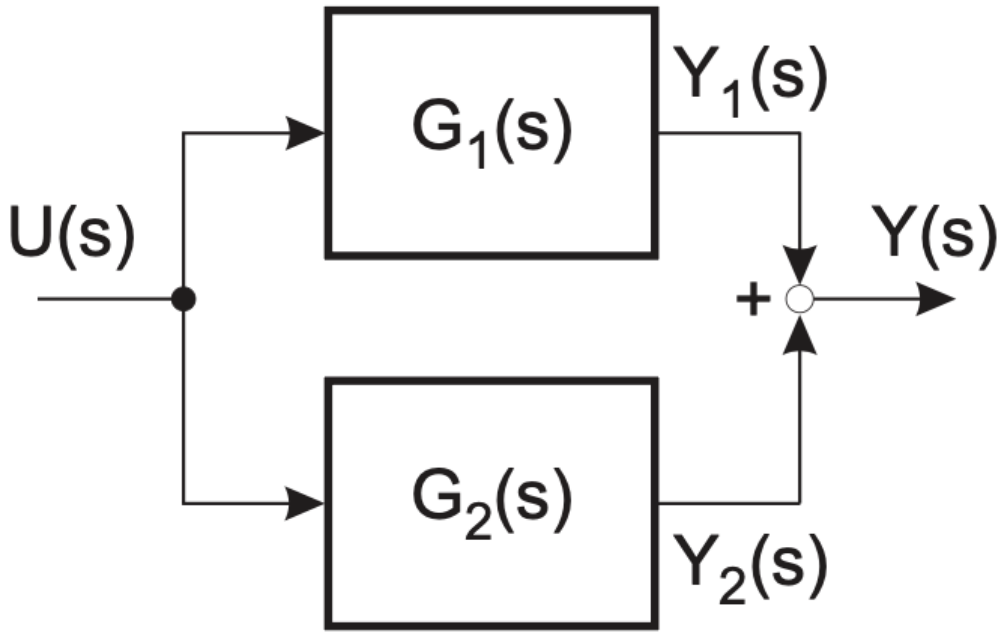


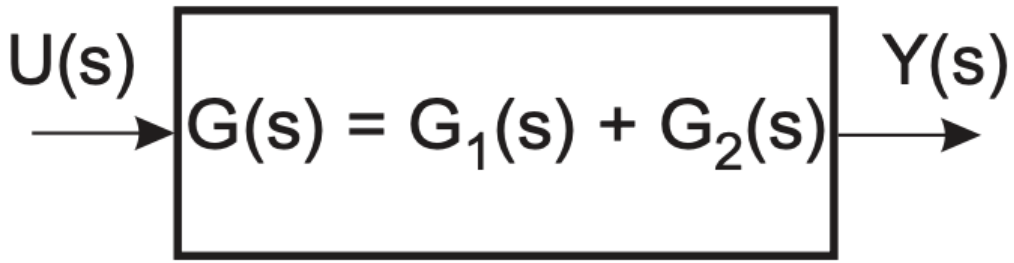
Series Connection

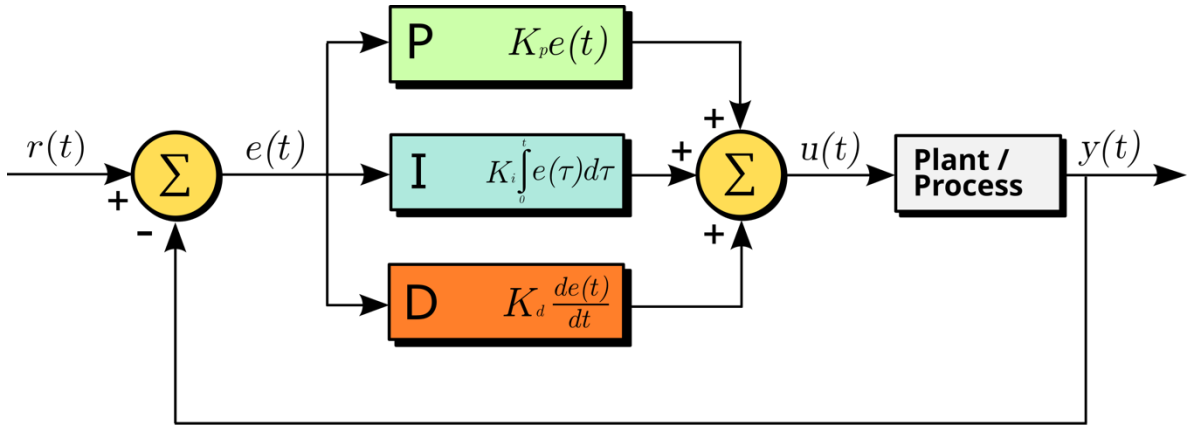




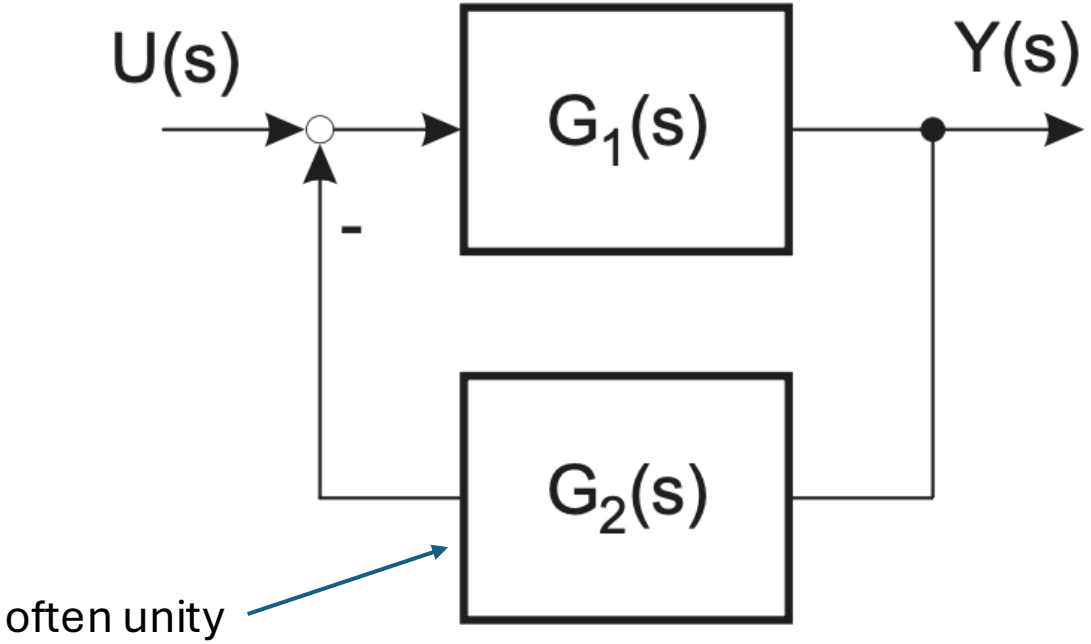
Parallel Connection







Feedback Connection

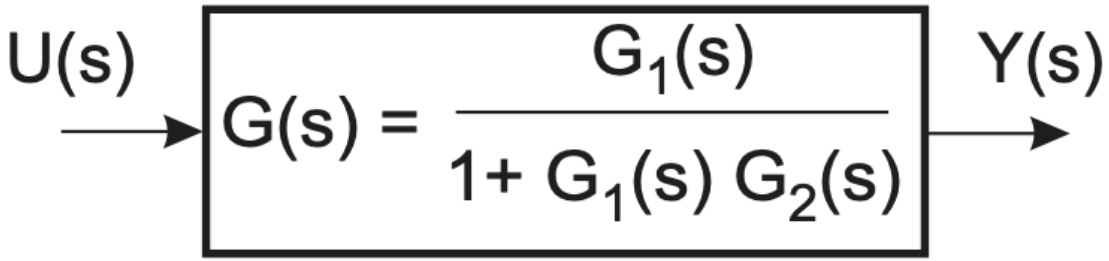


$$Y(s) = G_1(s) \cdot E(s)$$
$$E(s) = U(s) - G_2(s) \cdot Y(s)$$

$$\Rightarrow Y(s) = G_1(s) \cdot \big(U(s) - G_2(s) \cdot Y(s)\big)$$

$$\Rightarrow Y(s) \cdot \big(1 + G_1(s) \cdot G_2\,(s)\big) = G_1(s) \cdot U(s)$$

$$\Rightarrow \frac{Y(s)}{U(s)} = \frac{G_1(s)}{\big(1 + G_1(s) \cdot G_2\,(s)\big)}$$

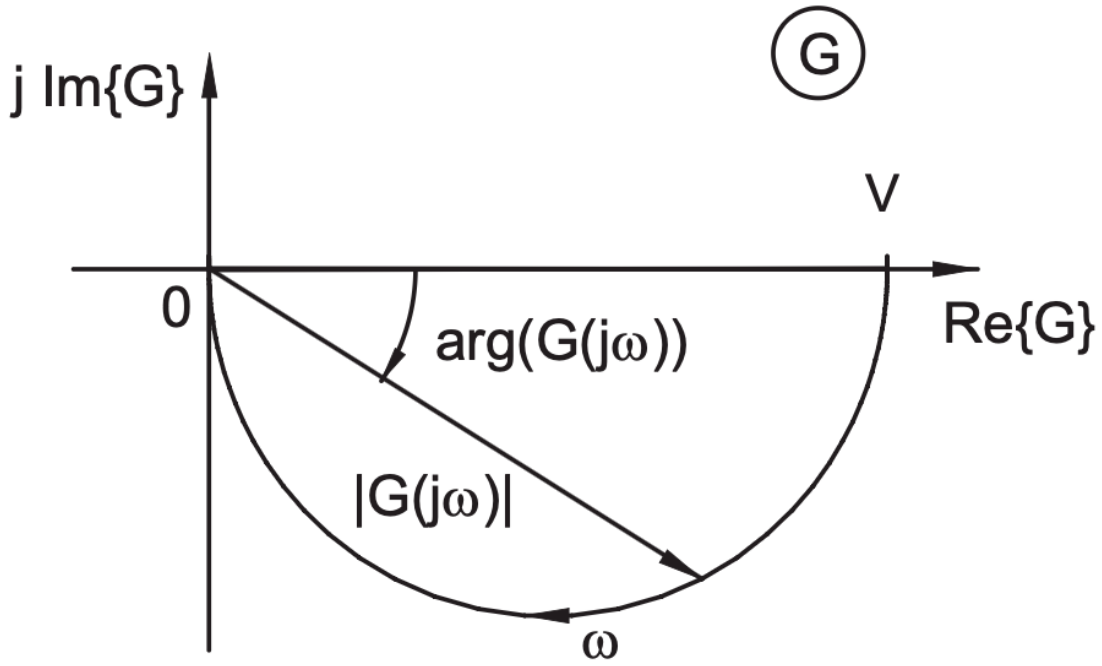


Response = Transient part + Steady-state part

Frequency Response

- Response to sinusoidal inputs: $G(j\omega)$
- Amplitude and phase as function of frequency ω

Nyquist diagram:



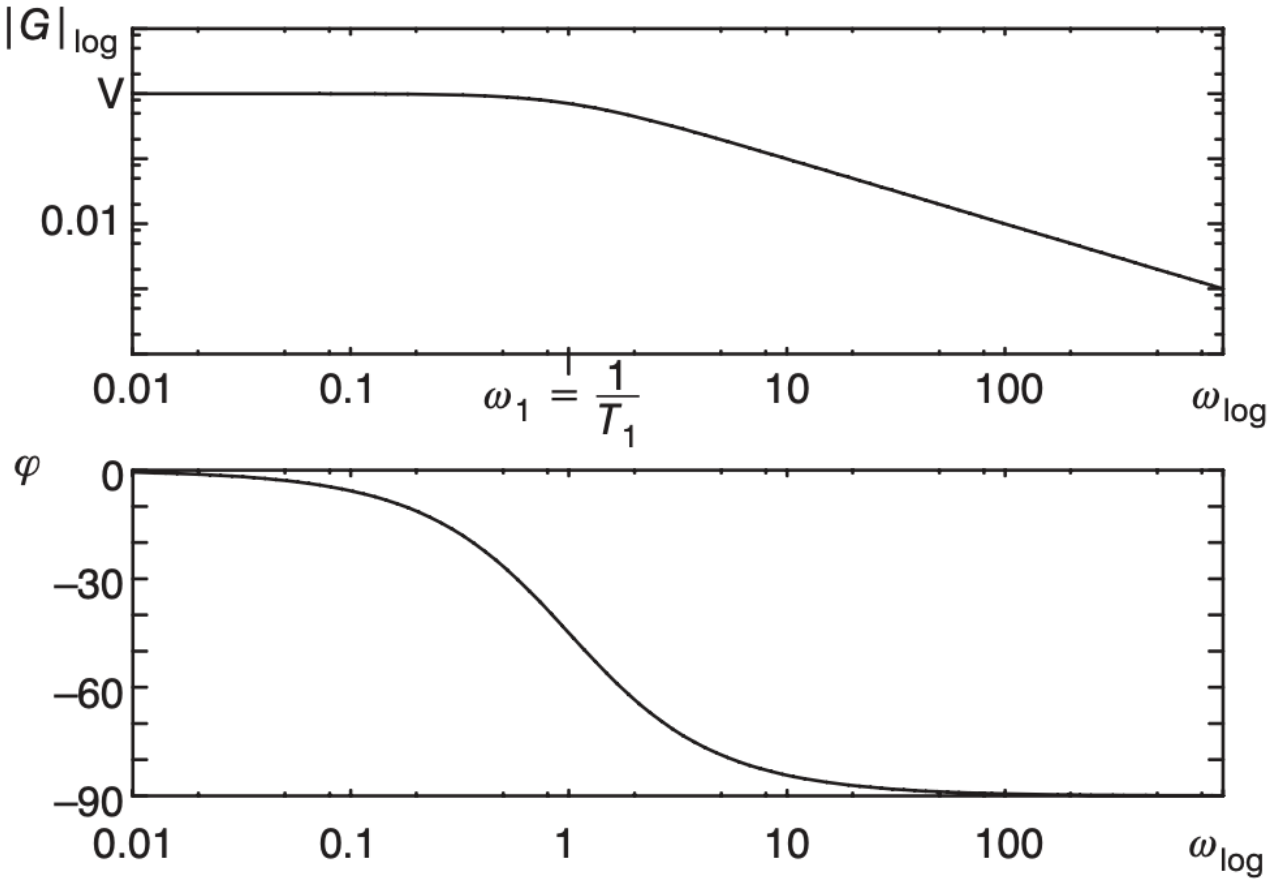
Bode Plots

1. Magnitude (dB) vs frequency

2. Phase vs frequency

Logarithmic frequency axis

Example: PT1



$$\mathrm{dB} = 10 \cdot \log_{10}\left(\frac{P_2}{P_1}\right)$$

For amplitudes ($P \sim A^2$):

$$\mathrm{dB} = 20 \cdot \log_{10}(|G(j\omega)|)$$

Bandwidth

Frequency range with good tracking (i.e., closed-loop system follows reference signal closely in shape, speed, and steady-state value)

Strong damping for higher frequencies (kind of filter on reference)

Typically consider frequency for which $|G(j\omega)|$ changes by -3 dB:
$|G(j\omega_B)| = \frac{1}{\sqrt{2}} \cdot |G(0)|$ (amplitude drops to 70.7% of its DC value)

Indicates speed of system: higher bandwidth → faster response

But also less robust and higher noise amplification

Individual Elements of Control Loops

Polynomial form of differential equation / transfer function:

$$(a_n s^n + \cdots + a_1 s + a_0) \cdot Y(s) = (b_m s^m + \cdots + b_1 s + b_0) \cdot U(s)$$

→ Transfer functions of individual elements by setting a portion of the coefficients to zero

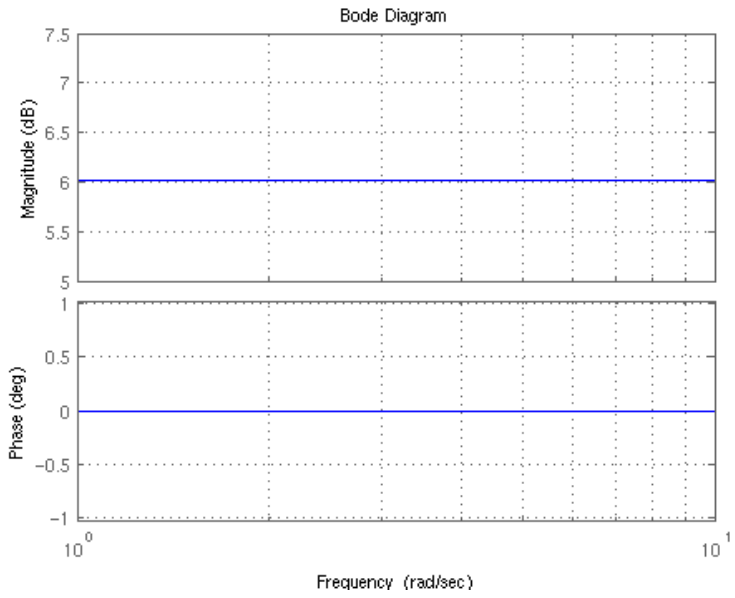
Proportional Element (P)

Consider $n = 0$:

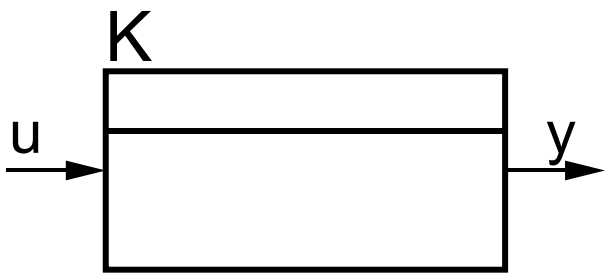
$$y(t) = \frac{b_0}{a_0} \cdot u(t) \quad \text{or} \quad Y(s) = \frac{b_0}{a_0} \cdot U(s)$$

$$\frac{b_0}{a_0} = K_P$$

(just constant gain)



Step response:



Integral Element (I)

Pole at $s = 0$

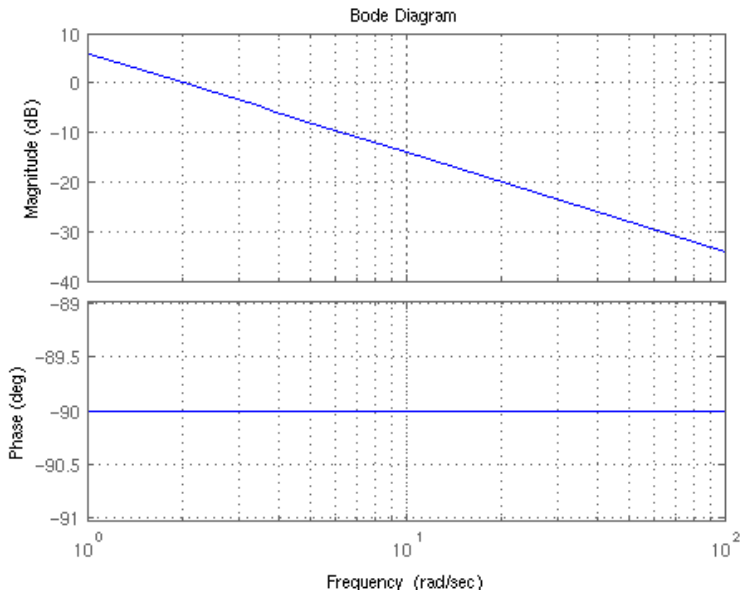
Consider $n = 1$, with $a_0 = b_1 = 0$:

$$\frac{dy(t)}{dt} = \frac{b_0}{a_1} \cdot u(t) \quad \text{or} \quad Y(s) = \frac{b_0}{a_1} \cdot \frac{1}{s} \cdot U(s)$$

$$y(t) = \frac{b_0}{a_1} \cdot \int u(t)\, dt$$

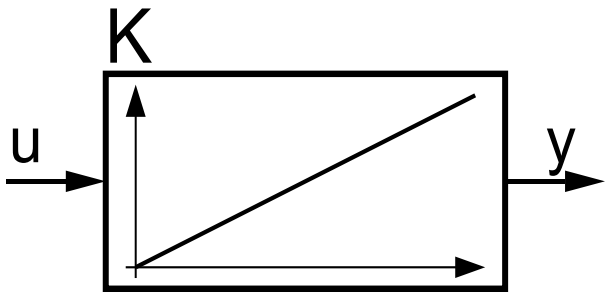
$$\frac{b_0}{a_1} = K_I = {}^{1}/_{T_I}$$

(integrator gain or integral time constant)



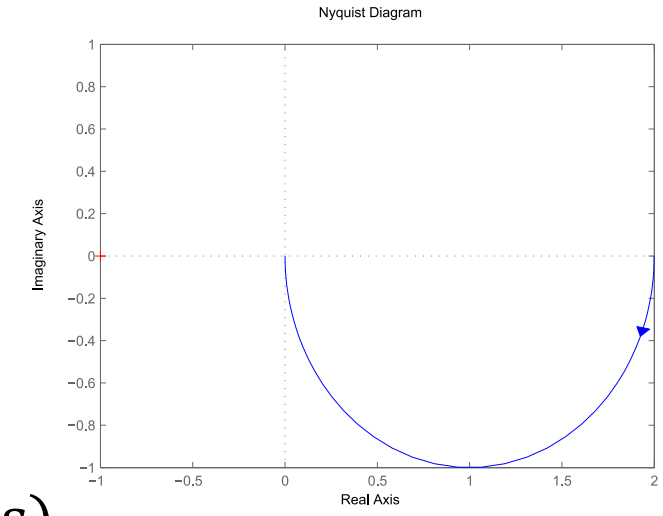
Step response:

$$\int \text{const}\, dt = t$$



Energy storage, memory

First-Order Lag Element (PT1)



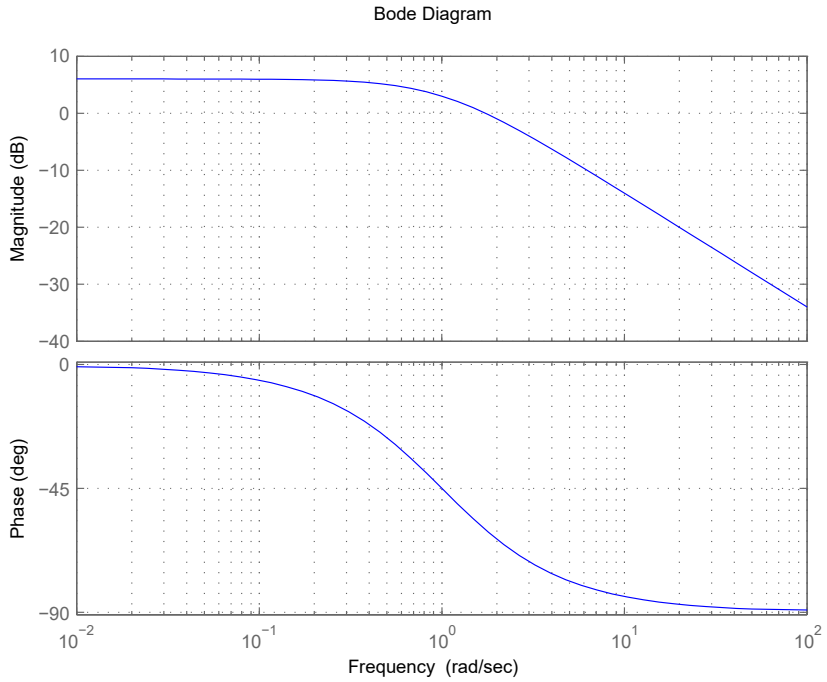
Consider $n = 1$, with $b_1 = 0$:

$$\frac{a_1}{a_0} \cdot \frac{dy(t)}{dt} + y(t) = \frac{b_0}{a_0} \cdot u(t) \quad \text{or} \quad Y(s) \cdot \left(s \cdot \frac{a_1}{a_0} + 1\right) = \frac{b_0}{a_0} \cdot U(s)$$

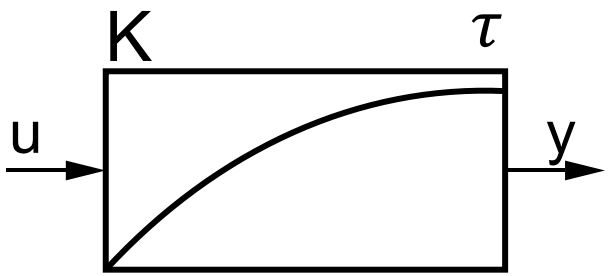
τ $\qquad$ K

$$\boxed{G(s) = \frac{Y(s)}{U(s)} = \frac{K}{1 + \tau s}}$$

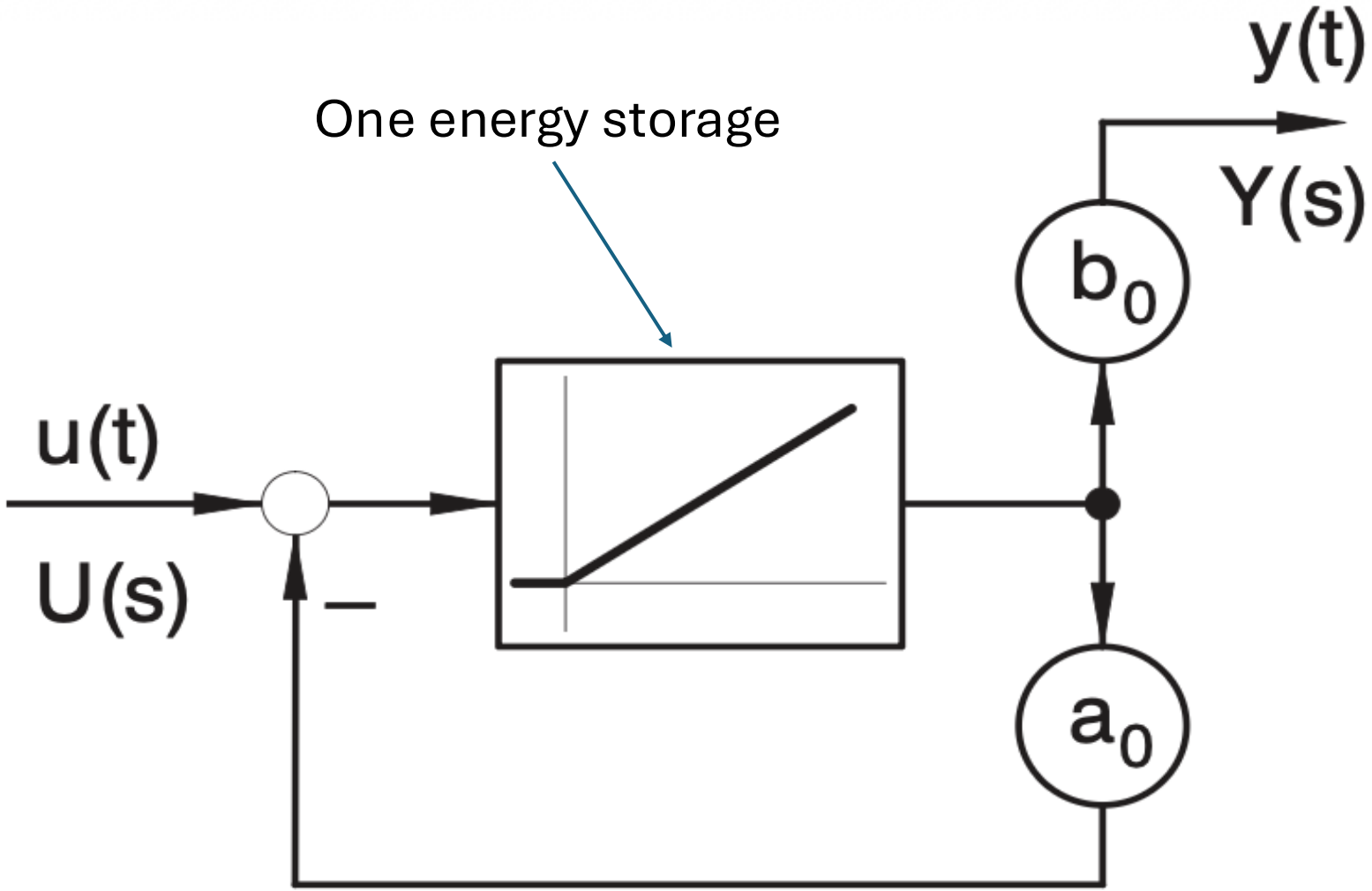
Pole at $s = -\frac{1}{\tau}$



Step response:



PT1 Signal Flow Diagram



Derivative Element (D)

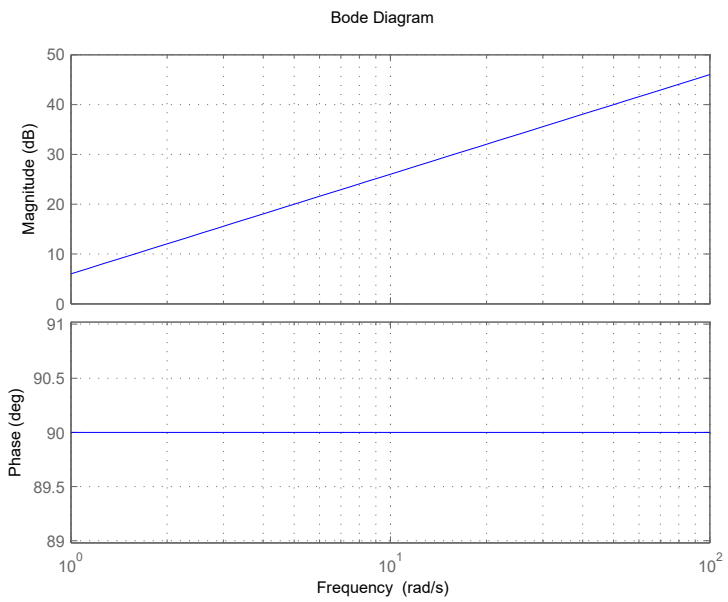
Amplifies high frequencies, i.e., noise
→ use filtered differentiator (DT1)

Consider $n = 1$, with $a_1 = b_0 = 0$:

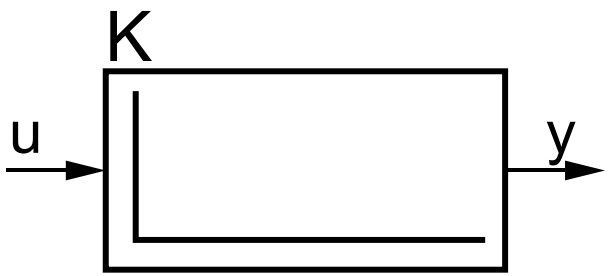
$$y(t) = \frac{b_1}{a_0} \cdot \frac{du(t)}{dt} \quad \text{or} \quad Y(s) = \frac{b_1}{a_0} \cdot s \cdot U(s)$$

$$\frac{b_1}{a_0} = K_D = T_D$$

(differentiator/derivative gain or derivative time constant)



Step response:



physically unrealizable → damping

Filtered Differentiator (DT1)

Used in PID controllers

Consider $n = 1$, with $b_0 = 0$:

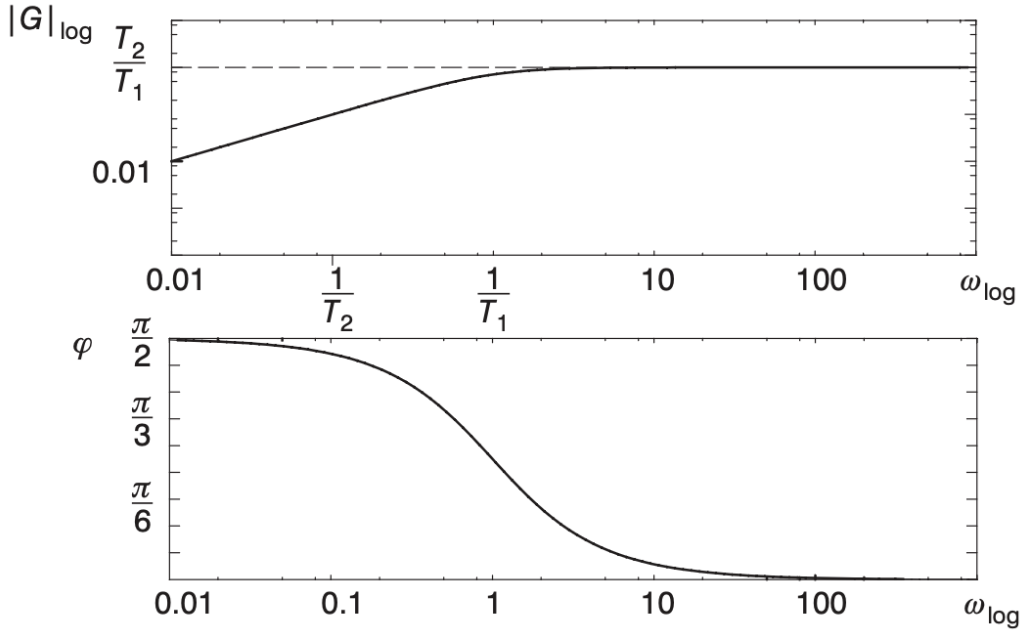
$$\frac{a_1}{a_0} \cdot \frac{dy(t)}{dt} + y(t) = \frac{b_1}{a_0} \cdot \frac{du(t)}{dt} \quad \text{or} \quad Y(s) \cdot \left(s \cdot \frac{a_1}{a_0} + 1\right) = \frac{b_1}{a_0} \cdot s \cdot U(s)$$

T_1 T_2

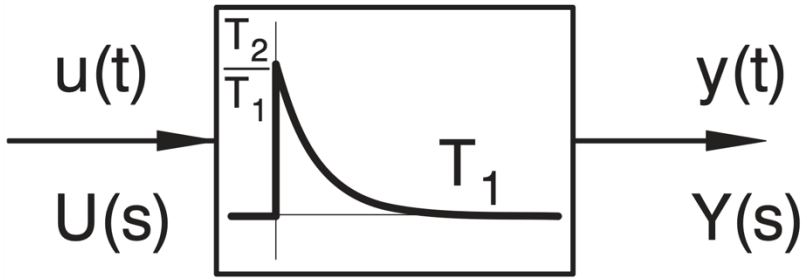
$$\boxed{G(s) = \frac{T_2 \cdot s}{T_1 \cdot s + 1}}$$

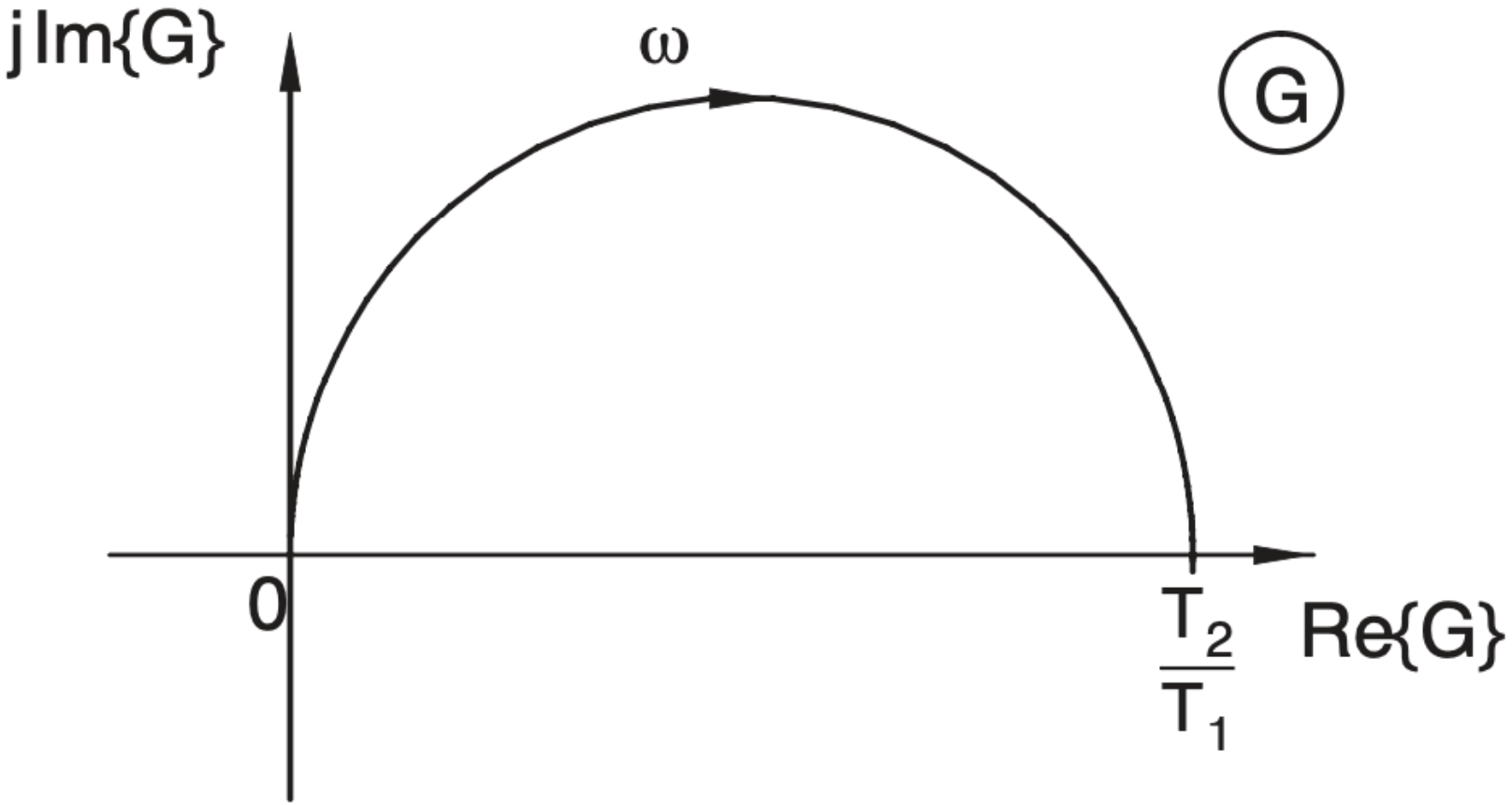
$$\lim_{T_1 \to 0} G(s) = T_2 \cdot s$$

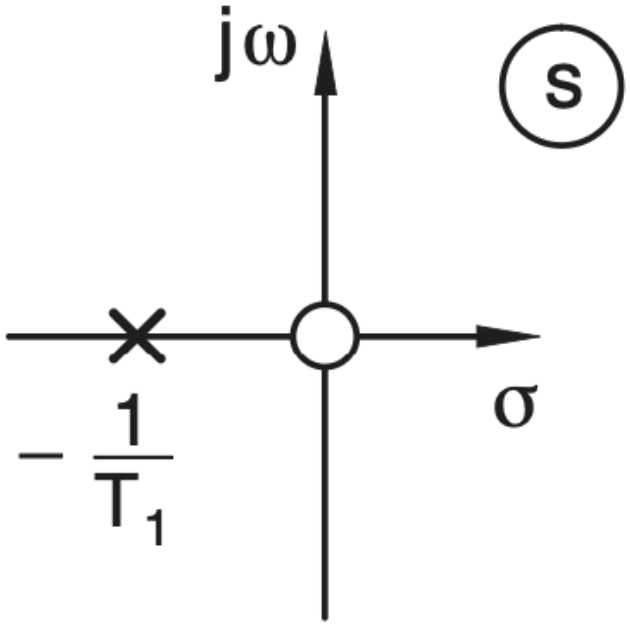
→ ideal D element



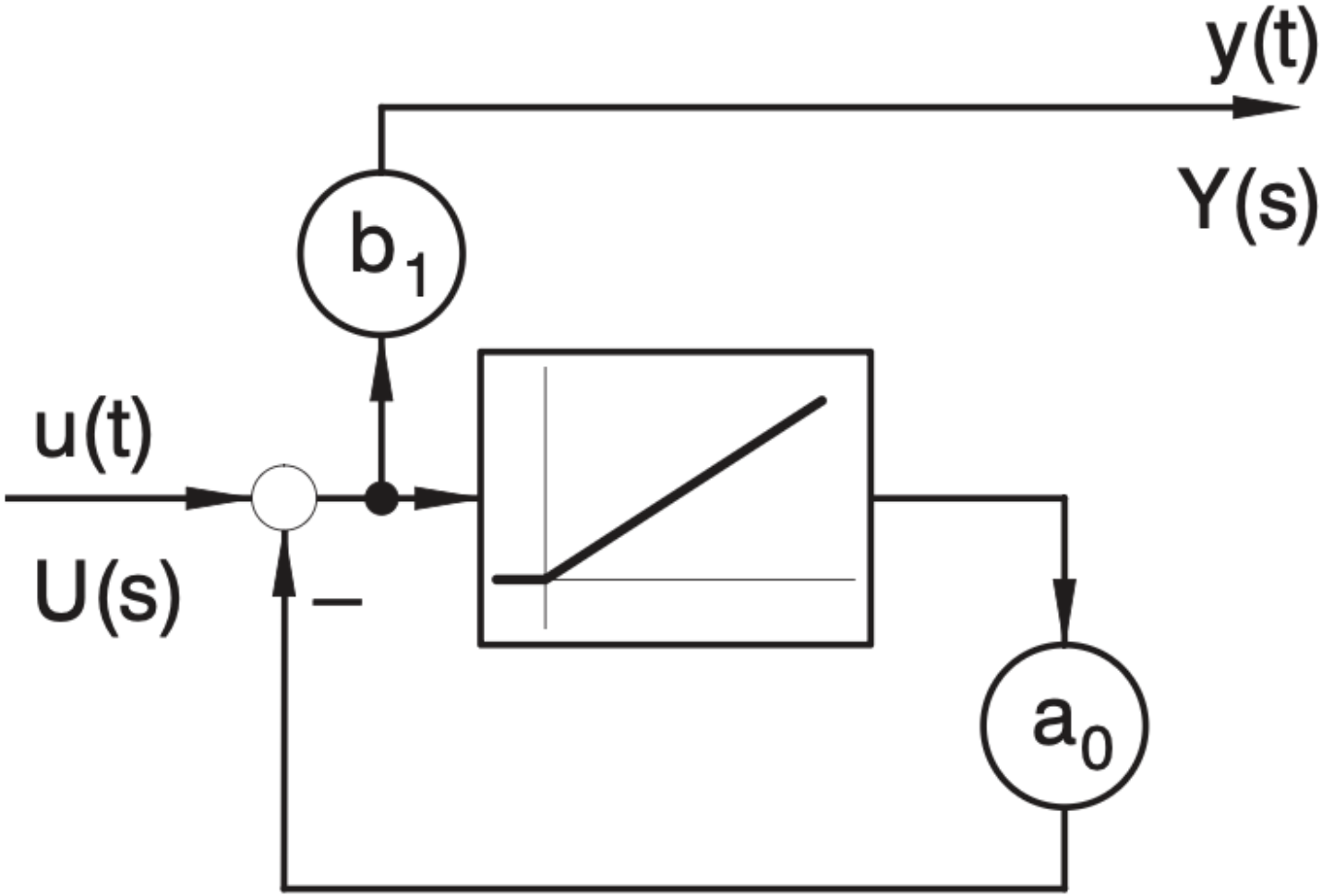
Step response:




jIm{G}
ω
G
0
T2/T1
Re{G}


jω
s
σ
− 1/T1

DT1 Signal Flow Diagram



Second-Order Lag Element (PT2)

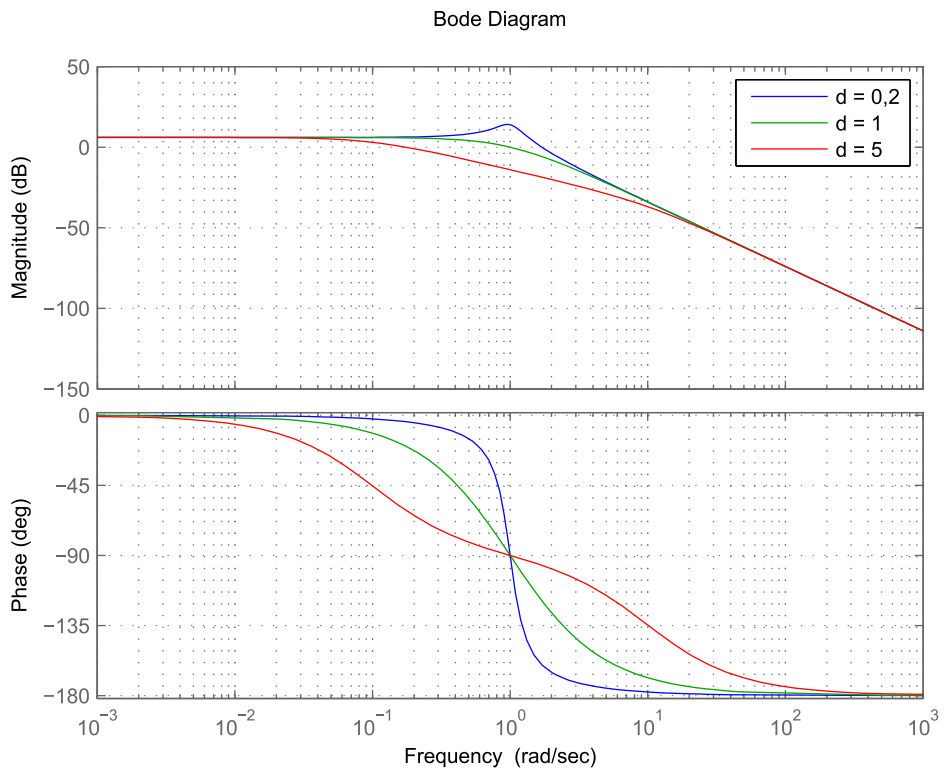
Consider $n = 2$, with $b_1 = b_2 = 0$:

$$\frac{a_2}{a_0}\frac{d^2y(t)}{dt^2} + \frac{a_1}{a_0}\frac{dy(t)}{dt} + y(t) = \frac{b_0}{a_0}u(t)$$

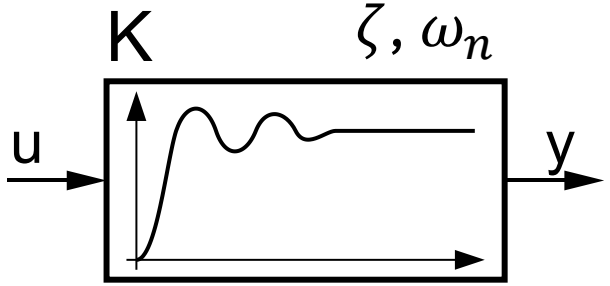
$$\frac{1}{\omega_0^2} \qquad \frac{2\zeta}{\omega_n} \qquad K$$

$$Y(s) \cdot \left(\frac{a_2}{a_0}s^2 + \frac{a_1}{a_0}s + 1\right) = \frac{b_0}{a_0} \cdot U(s)$$

$$\boxed{G(s) = \frac{K}{\frac{1}{\omega_n^2}s^2 + \frac{2\zeta}{\omega_n}s + 1}}$$



Step response:



PT2 as Product of Two PT1 Elements

$G(s) = \frac{K}{\frac{1}{\omega_n^2}s^2 + \frac{2\zeta}{\omega_n}s + 1}$ can be written as product of two PT1 elements:

$$G(s) = \frac{K}{(\tau_1 s + 1)(\tau_2 s + 1)}$$

individual time constants

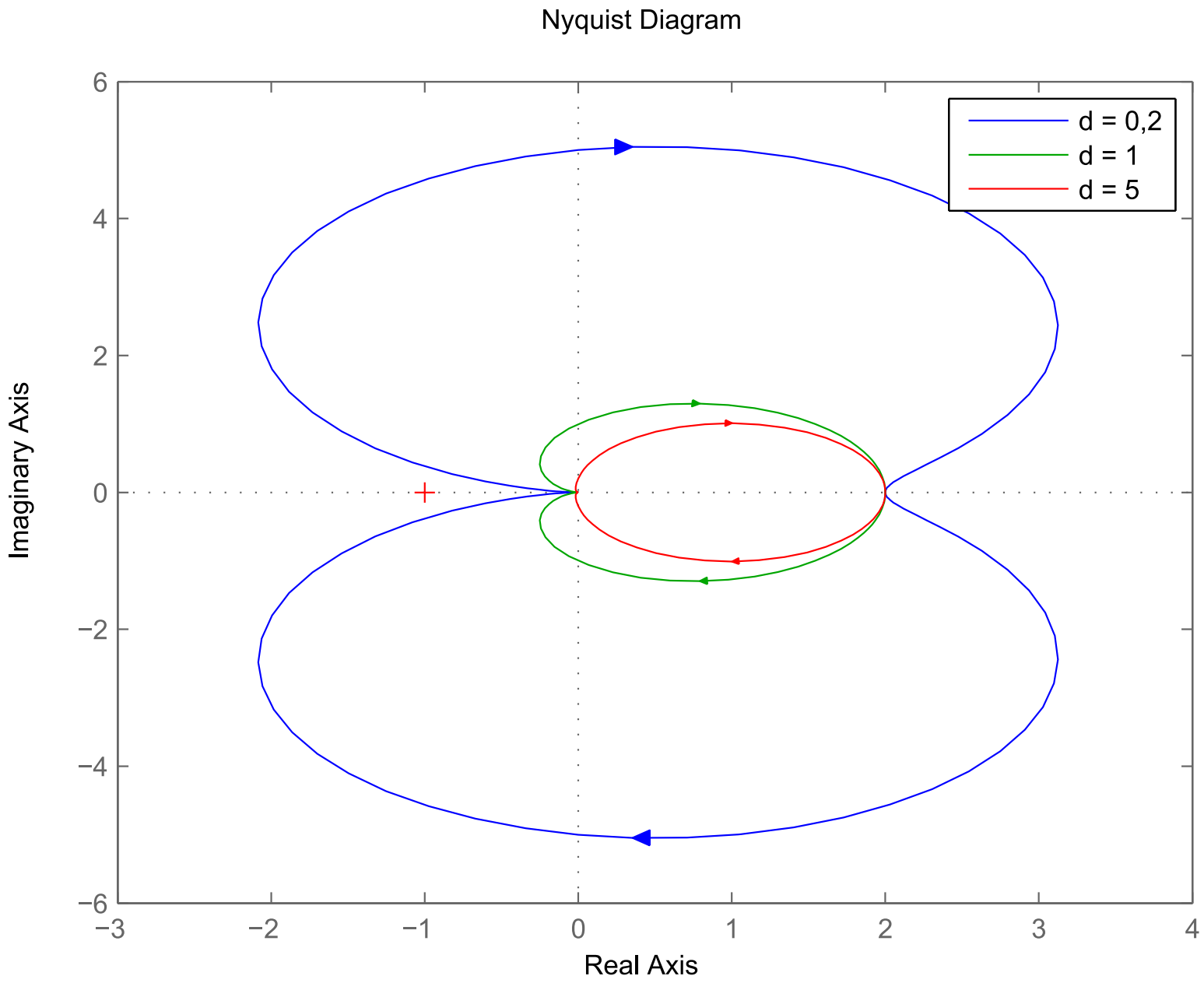
$$\Rightarrow \frac{1}{\omega_n^2}s^2 + \frac{2\zeta}{\omega_n}s + 1 = \tau_1\tau_2 s^2 + (\tau_1 + \tau_2)s + 1$$

$$\Rightarrow \frac{1}{\omega_n^2} = \tau_1\tau_2, \quad \frac{2\zeta}{\omega_n} = (\tau_1 + \tau_2)$$

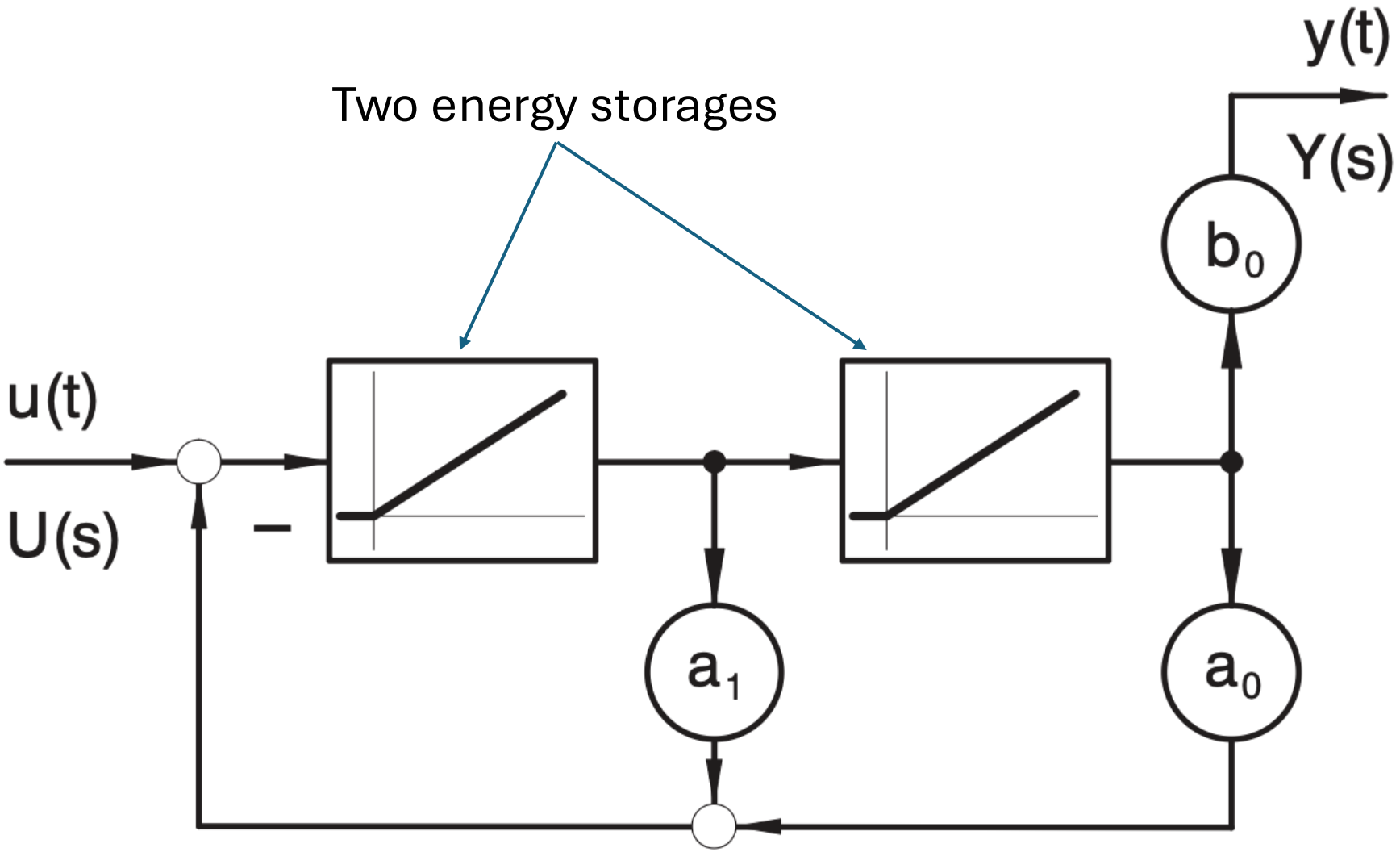
$$\Rightarrow \zeta = \frac{\tau_1 + \tau_2}{2\sqrt{\tau_1\tau_2}}$$

for real PT1 elements: $\tau_1 + \tau_2 \geq 2\sqrt{\tau_1\tau_2}$ (arithmetic-geometric mean inequality)

→ overdamped ($\zeta < 1$) or critically damped ($\zeta = 1$) for $\underbrace{\tau_1 = \tau_2}_{\text{fastest non-oscillatory response}}$

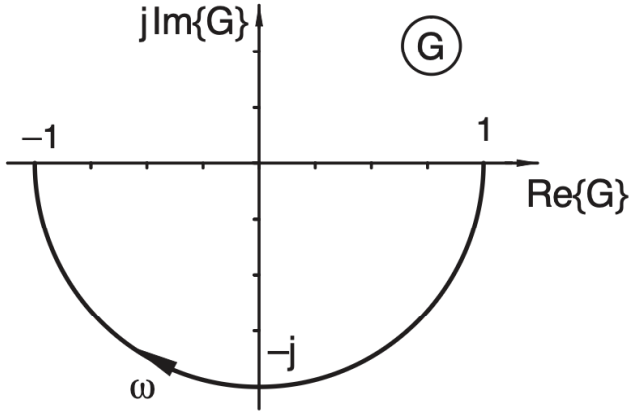

Nyquist Diagram
d = 0,2
d = 1
d = 5
Imaginary Axis
Real Axis
6
4
2
0
−2
−4
−6
−3
−2
−1
0
1
2
3
4

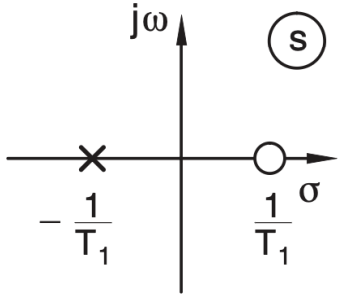
PT2 Signal Flow Diagram



All-Pass Systems

first-order all-pass: $G(s) = \frac{s-a}{s+a}$





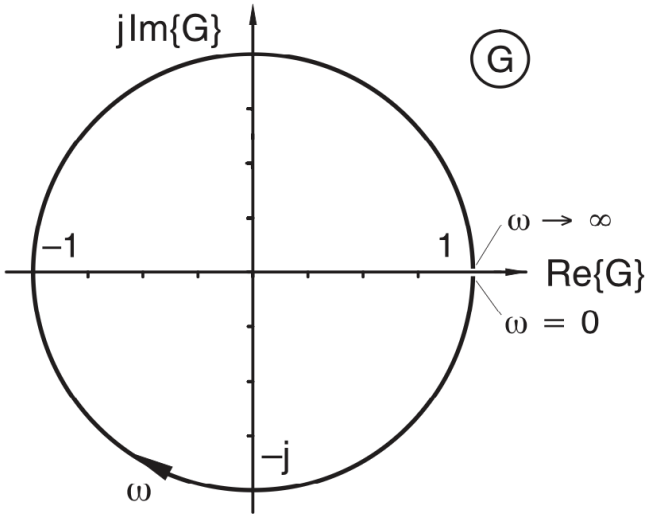
Magnitude constant for all frequencies ($|G(j\omega)| = 1$), but phase depends on frequency

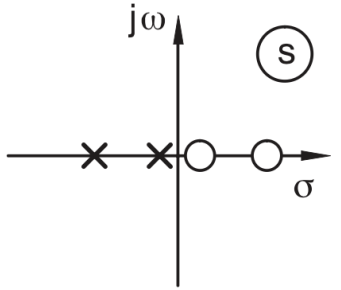
coefficients: $b_k = (-1)^{n-k} a_{n-k}$

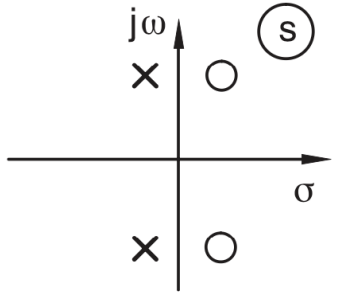
→ Zeros are mirror images of the poles with respect to the imaginary axis.

Important in signal processing, e.g., for runtime alignment

second-order all-pass: $G(s) = \frac{a_2 s^2 - a_1 s + a_0}{a_2 s^2 + a_1 s + a_0}$







Systems with Dead Time

The transfer function of a system can be extended by multiplying it with the exponential term that represents a time delay:

$$G(s) = G_0(s) \cdot e^{-s \cdot \tau_D}$$

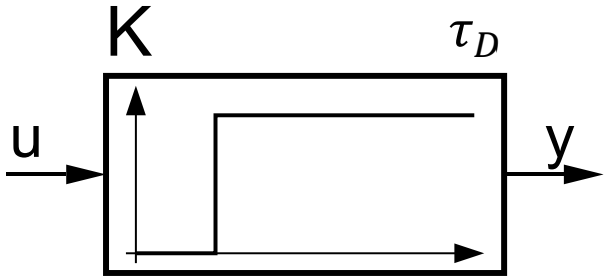
original transfer function

time delay (aka dead time)

Example: PT1 with dead time

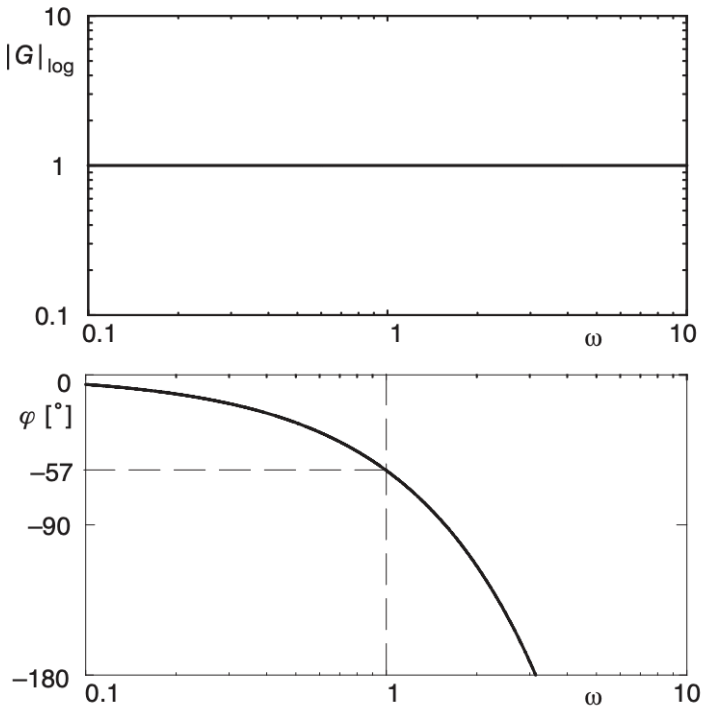
$$G(s) = \frac{K}{1+\tau \cdot s} \cdot e^{-s \cdot \tau_D}$$

Step response (P element with dead time):

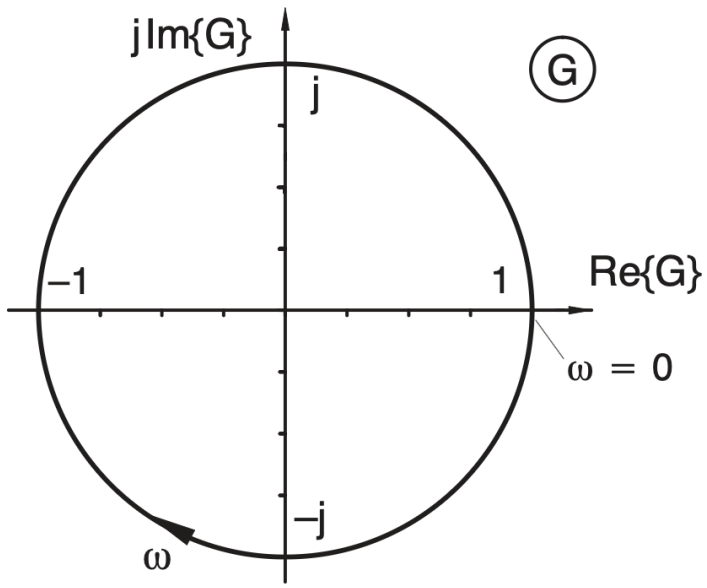


Mainly suitable for frequency response analysis (e.g., Bode plots)

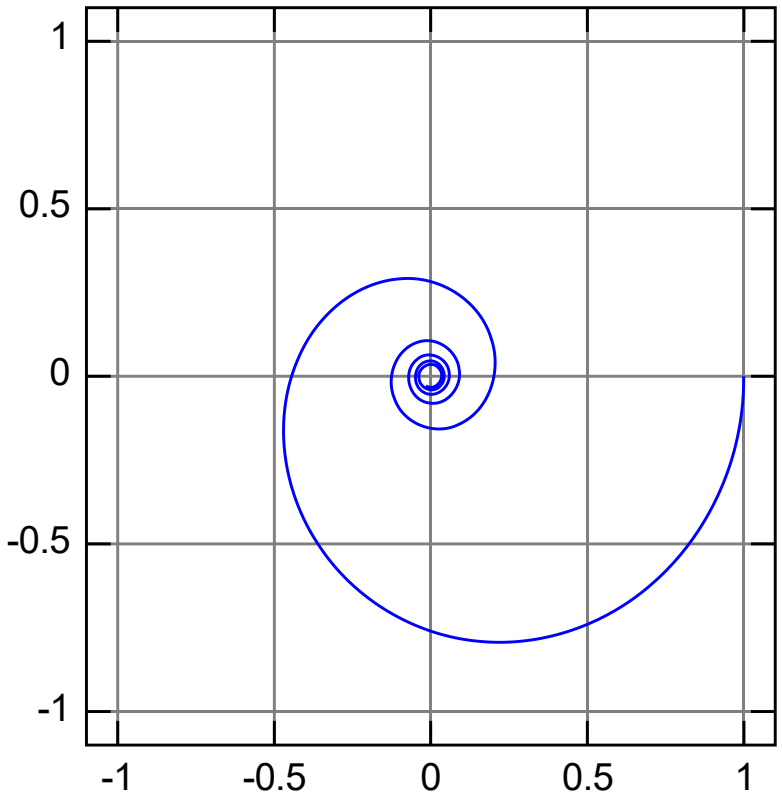
Bode diagram for system with dead time:



Nyquist diagram for system with dead time:



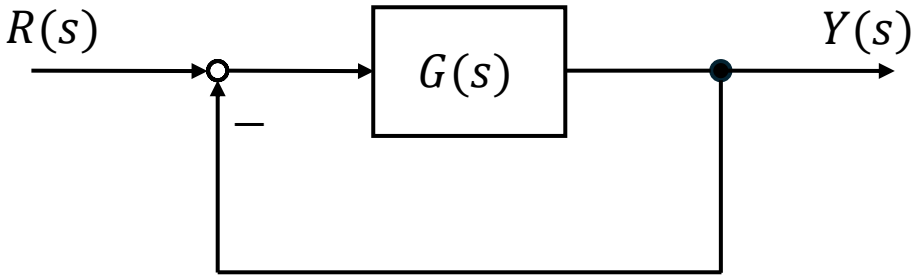
Nyquist diagram for PT1 with dead time:



Control Loop

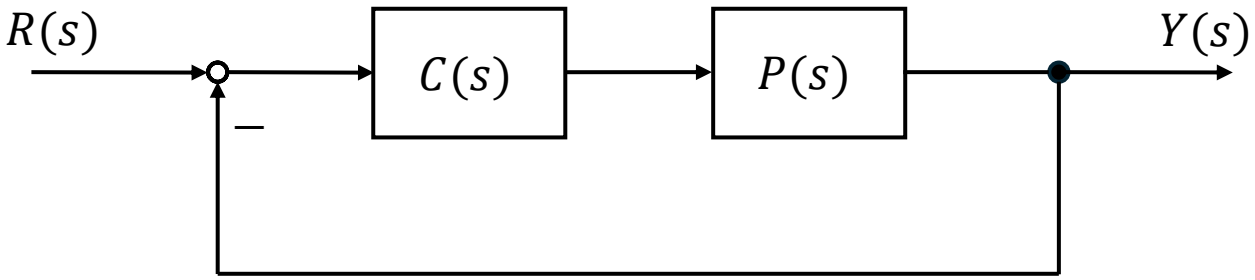
$$Y(s) = G(s) \cdot (R(s) - Y(s))$$

$$\Rightarrow Y(s) \cdot (1 + G(s)) = G(s) \cdot R(s)$$



$$\Rightarrow \frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$

closed-loop transfer function for negative feedback



$$\frac{Y(s)}{R(s)} = \frac{C(s) \cdot P(s)}{1 + C(s) \cdot P(s)}$$

$G(s)$ above combines controller $C(s)$ and plant $P(s)$

PID Controller Transfer Function

$$C(s) = K_p + \frac{K_i}{s} + K_d \cdot s$$

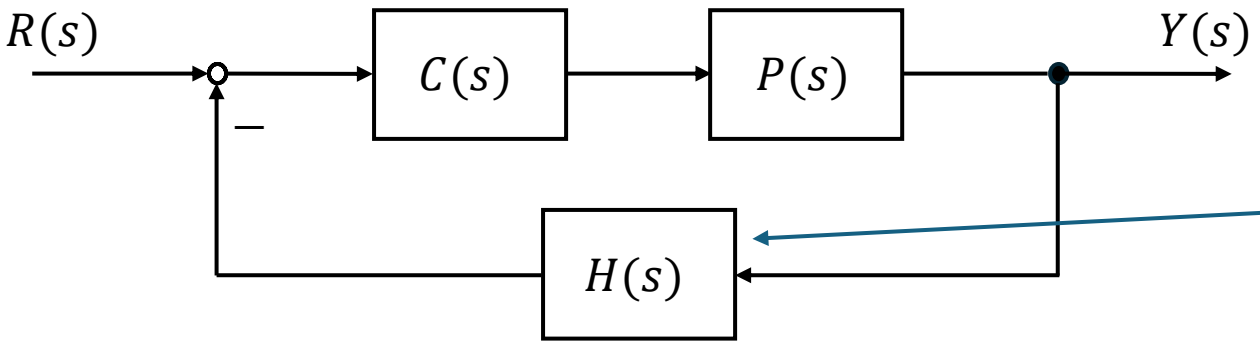
- K_p: increases loop gain → faster response, smaller steady-state error, may increase overshoot
- K_i: adds integrator → eliminates steady-state error, may reduce stability margin
- K_d: adds damping → reduces overshoot, improves transient behavior

Controller design:

- Shape the closed-loop poles to obtain the desired dynamics
- Achieved by selecting and tuning controller parameters (move the closed-loop poles)

Hidden Assumptions

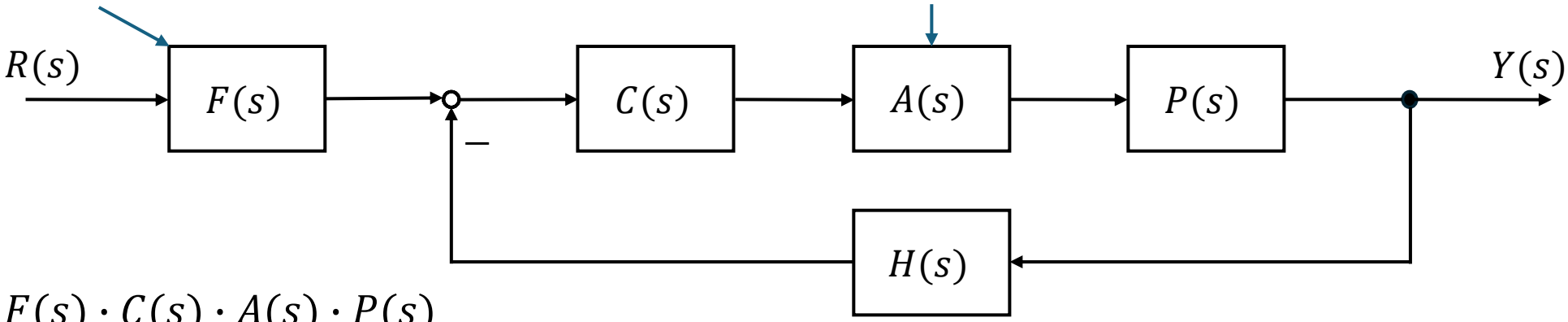
$$\frac{Y(s)}{R(s)} = \frac{C(s) \cdot P(s)}{1 + C(s) \cdot P(s) \cdot H(s)}$$



sensor dynamics: If sensor is fast, one can assume unity feedback.

reference prefilter: If reference shaping doesn't affect stability, one can treat it separately.

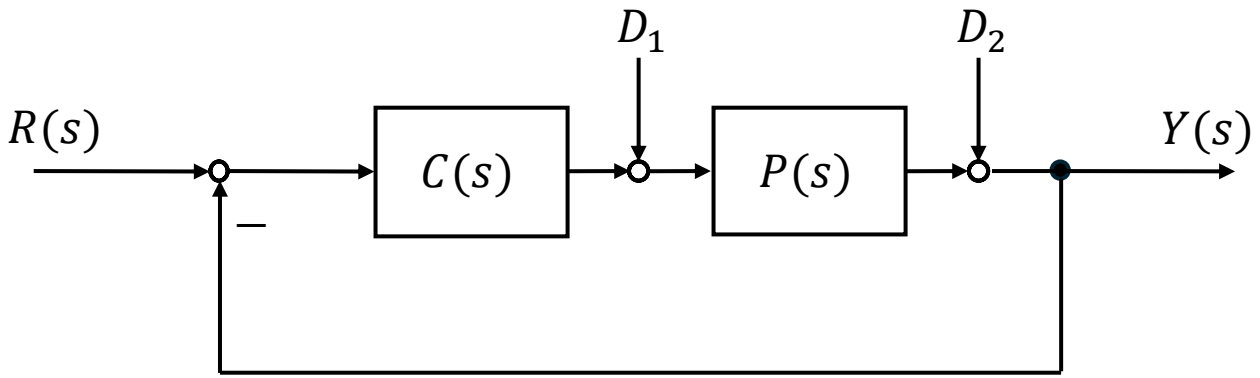
actuator dynamics: If actuator bandwidth ≫ closed-loop bandwidth, one can ignore it.



$$\frac{Y(s)}{R(s)} = \frac{F(s) \cdot C(s) \cdot A(s) \cdot P(s)}{1 + C(s) \cdot A(s) \cdot P(s) \cdot H(s)}$$

Always model what can affect poles

Disturbances



$$Y(s) = \frac{C(s) \cdot P(s)}{1 + C(s) \cdot P(s)} \cdot R(s) + \frac{P(s)}{1 + C(s) \cdot P(s)} \cdot D_1(s) + \frac{1}{1 + C(s) \cdot P(s)} \cdot D_2(s)$$

Output as superposition of input and disturbances